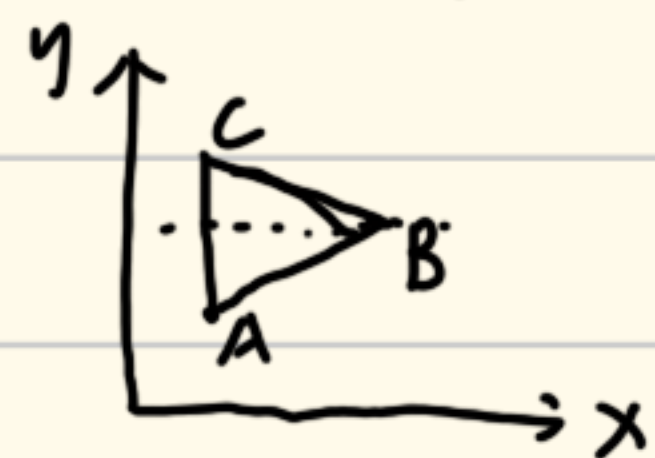


2400015490 冯浩杰

8.3 1. (3) $\oint_L \sqrt{x^2+y^2} dx + y[xy + \ln(x+\sqrt{x^2+y^2})] dy$, $L: A(1,1), B(2,2), C(1,3)$



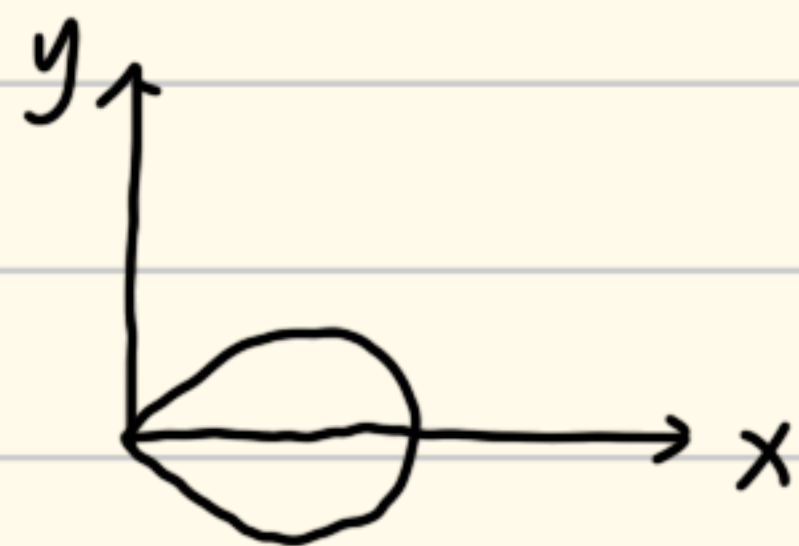
$$P = \sqrt{x^2+y^2}, \quad Q = y[xy + \ln(x+\sqrt{x^2+y^2})]$$

$$\frac{\partial Q}{\partial x} = y^2 + \frac{1 + \frac{x}{\sqrt{x^2+y^2}}}{x + \sqrt{x^2+y^2}} = y^2 + \frac{x + \sqrt{x^2+y^2}}{\sqrt{x^2+y^2}(x + \sqrt{x^2+y^2})} = y^2 + \frac{y}{\sqrt{x^2+y^2}}$$

$$\frac{\partial P}{\partial y} = \frac{y}{\sqrt{x^2+y^2}}$$

$$\begin{aligned} \text{原式} &= \iint_D y^2 d\sigma = \int_1^2 dy \int_1^y y^2 dx + \int_2^3 dy \int_1^{4-y} y^2 dx \\ &= \int_1^2 y^2(y-1) dy + \int_2^3 (3-y) y^2 dy \\ &= \int_1^2 (y^3 - y^2) dy + \int_2^3 (3y^2 - y^3) dy \\ &= \frac{15}{4} - \frac{7}{3} + 19 - \frac{65}{4} \\ &= \frac{50}{3} - \frac{25}{2} \\ &= \frac{25}{6} \end{aligned}$$

(4) $\oint_{L^+} (x + e^x \sin y) dx + (x + e^x \cos y) dy$, 其中 L 是双纽线 $r^2 = \cos 2\theta$ 的右半支;



$$P = x + e^x \sin y, \quad Q = x + e^x \cos y$$

$$\frac{\partial P}{\partial y} = e^x \cos y, \quad \frac{\partial Q}{\partial x} = 1 + e^x \cos y$$

$$\begin{aligned} \therefore \text{原式} &= \iint_D \partial \sigma = 2 \cdot \int_0^{\frac{\pi}{4}} d\theta \int_0^{\sqrt{\cos 2\theta}} r \cdot dr \\ &= \int_0^{\frac{\pi}{4}} \cos 2\theta \cdot d\theta \\ &= \frac{1}{2} \sin 2\theta \Big|_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \end{aligned}$$

(2) 心脏线

$$\begin{cases} x = a(1 - \cos t) \cos t, \\ y = a(1 - \cos t) \sin t, \end{cases} \quad 0 \leq t \leq 2\pi.$$

$$S_D = \frac{1}{2} \oint_L -y dx + x dy,$$

$$dx = (a \sin t \cos t - a(1 - \cos t) \sin t) dt = (2a \sin t \cos t - a \sin t) dt = a(\sin 2t - \sin t) dt$$

$$dy = (a \sin^2 t + a(1 - \cos t) \cos t) dt = a(\sin^2 t - \cos^2 t + \cos t) dt = a(-\cos 2t + 1 + \cos t) dt$$

$$\oint_L -y dx + x dy = \int_0^{2\pi} a^2 [(1 - \cos t) \cos t (-\cos 2t + 1 + \cos t) - (1 - \cos t) \sin t (\sin 2t - \sin t)] dt$$

$$= \int_0^{2\pi} a^2 (1 - \cos t) [-\cos t \cos 2t + \cos^2 t - \sin t \sin 2t + \sin^2 t] dt$$

$$= \int_0^{2\pi} a^2 (1 - \cos t) (1 - \cos t) dt$$

$$= a^2 \int_0^{2\pi} (\cos^2 t - 2\cos t + 1) dt$$

$$= 3\pi a^2$$

$$\therefore S_D = \frac{3}{2} \pi a^2$$

(3) $\int_{(0,0)}^{(a,b)} e^x \cos y dx - e^x \sin y dy.$

$$\frac{\partial P}{\partial y} = -e^x \sin y, \quad \frac{\partial Q}{\partial x} = -e^x \sin y, \quad \text{恒相等}$$

则与路径无关

$$\frac{\partial u}{\partial x} = e^x \cos y, \quad \text{则 } u(x,y) = e^x \cos y + \varphi(y), \quad \frac{\partial u}{\partial y} = -e^x \sin y + \varphi'(y) = -e^x \sin y$$

$$\text{则 } u = e^x \cos y, \quad \text{原式} = u(a,b) - u(0,0) = e^a \cos b - 1$$

6. 求满足下列等式的函数 $u(x,y)$:

(1) $du = (x^2 + 2xy - y^2) dx + (x^2 - 2xy - y^2) dy;$

(2) $du = (2x \cos y - y^2 \sin x) dx + (2y \cos x - x^2 \sin y) dy.$

$$\frac{\partial u}{\partial x} = 2x \cos y - y^2 \sin x, \quad u(x,y) = x^2 \cos y + y^2 \cos x + \varphi(y)$$

$$\frac{\partial u}{\partial y} = -x^2 \sin y + 2y \cos x + \varphi'(y) = 2y \cos x - x^2 \sin y$$

$$\varphi'(y) = 0$$

$$\therefore u(x,y) = x^2 \cos y + y^2 \cos x + C$$

⑧ 求 $\int_{(0,1)}^{(1,1)} \left(\frac{x}{\sqrt{x^2+y^2}} + y \right) dx + \left(\frac{y}{\sqrt{x^2+y^2}} + x \right) dy$.

$$\frac{\partial P}{\partial y} = \frac{-x \cdot \frac{y}{\sqrt{x^2+y^2}}}{x^2+y^2} = \frac{-xy}{(x^2+y^2)^{\frac{3}{2}}} + 1, \quad \frac{\partial Q}{\partial x} = \frac{-xy}{(x^2+y^2)^{\frac{3}{2}}} + 1,$$

$$\frac{\partial P}{\partial y} \equiv \frac{\partial Q}{\partial x}, \text{ 有连续的一阶偏导}$$

则与路径无关

$$\frac{\partial u}{\partial x} = \frac{x}{\sqrt{x^2+y^2}} + y, \quad u(x, y) = \sqrt{x^2+y^2} + xy + \varphi(y),$$

$$\frac{\partial u}{\partial y} = \frac{y}{\sqrt{x^2+y^2}} + x + \varphi'(y), \quad \text{则 } \varphi'(y) = 0$$

$$\text{则 } u(x, y) = \sqrt{x^2+y^2} + xy,$$

$$\text{原式} = u(1, 1) - u(0, 1) = \sqrt{2} + 1 - 1 = \sqrt{2}$$

⑩ 设函数 $u(x, y), v(x, y)$ 在有界闭区域 D 上有连续的二阶偏导数, L 为 D 的边界, 分段光滑. 证明:

$$(1) \iint_D v \Delta u \, d\sigma = \oint_{L^+} v \frac{\partial u}{\partial n} \, ds - \iint_D \left(\frac{\partial u}{\partial x} \cdot \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \cdot \frac{\partial v}{\partial y} \right) d\sigma, \text{ 其中 } \frac{\partial u}{\partial n} \text{ 为 } u \text{ 沿 } L \text{ 的外法}$$

线方向的方向导数;

$$(2) \iint_D (u \Delta v - v \Delta u) \, d\sigma = \oint_{L^+} \left(u \frac{\partial v}{\partial n} - v \frac{\partial u}{\partial n} \right) ds.$$

(1) 设 L 的切方向单位向量为 $(\cos \alpha, \cos \beta)$, 则 $\vec{n} = (\cos \beta, -\cos \alpha)$

$$\oint_{L^+} v \frac{\partial u}{\partial n} \, ds = \oint_{L^+} v \cdot \left(\frac{\partial u}{\partial x} \cos \beta - \frac{\partial u}{\partial y} \cos \alpha \right) ds$$

$$= \oint_{L^+} -\frac{\partial u}{\partial y} v \, dx + \frac{\partial u}{\partial x} v \, dy$$

$$= \iint_D \left(\frac{\partial^2 u}{\partial x^2} v + \frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial^2 u}{\partial y^2} v + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right) d\sigma$$

$$= \iint_D v \cdot \Delta u \, d\sigma + \iint_D \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} \right) d\sigma$$

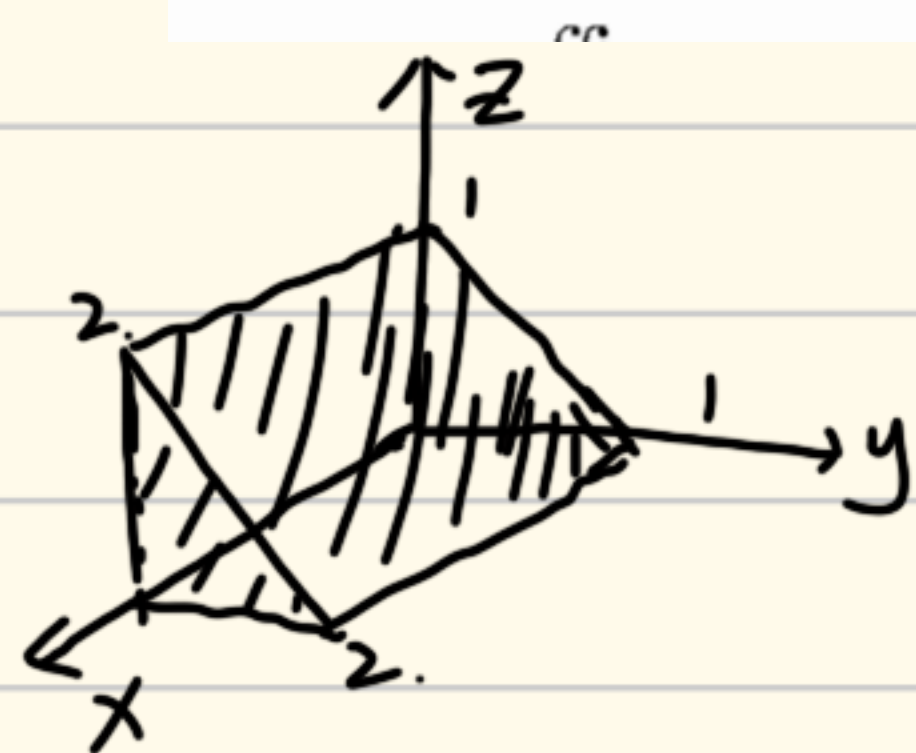
移项即得证

$$\begin{aligned}
(2) \quad & \int_{L^+} \left(u \frac{\partial v}{\partial \bar{n}} - v \frac{\partial u}{\partial \bar{n}} \right) ds \\
&= \int_{L^+} u \left(\frac{\partial v}{\partial x} \cos \beta - \frac{\partial v}{\partial y} \cos \alpha \right) - v \left(\frac{\partial u}{\partial x} \cos \beta - \frac{\partial u}{\partial y} \cos \alpha \right) ds \\
&= \int_{L^+} \left[\left(u \frac{\partial v}{\partial x} - v \frac{\partial u}{\partial x} \right) \cos \beta - \left(u \frac{\partial v}{\partial y} - v \frac{\partial u}{\partial y} \right) \cos \alpha \right] ds \\
&= \int_{L^+} \underbrace{\left(u \frac{\partial v}{\partial y} - v \frac{\partial u}{\partial y} \right)}_p \cdot dx + \underbrace{\left(u \frac{\partial v}{\partial x} - v \frac{\partial u}{\partial x} \right)}_q \cdot dy \\
&= \iiint_B \left(\frac{\partial u}{\partial x} \frac{\partial v}{\partial x} + u \frac{\partial^2 v}{\partial x^2} - \frac{\partial v}{\partial x} \frac{\partial u}{\partial x} - v \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} \frac{\partial v}{\partial y} + u \frac{\partial^2 v}{\partial y^2} - \frac{\partial v}{\partial y} \frac{\partial u}{\partial y} - v \frac{\partial^2 u}{\partial y^2} \right) d\Omega \\
&= \iiint_B (u \Delta v - v \Delta u) d\Omega
\end{aligned}$$

得证

8.4.

②. $\iint_S (y+z) dS$, 其中 S 为平面 $y+z=1$, $x=2$, 以及三个坐标平面所围成之立体的表面.



平面 1: $z=1-y$, $z_x=0$, $z_y=-1$

$$\begin{aligned}
& \iint_{S_1} (y+1-y) \cdot \sqrt{1+1} \cdot d\Omega \\
&= \iint_{S_1} \sqrt{2} d\Omega = 2\sqrt{2}
\end{aligned}$$

平面 2: $x=2$, $x_y=x_z=0$, $dS=1$

$$\iint_{S_2} (y+z) \cdot d\Omega = \int_0^1 dy \int_0^{1-y} (y+z) dz = \int_0^1 \left(y + \frac{1}{2}(1-y)^2 \right) dy = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$$

平面 3: $x=0$, 同上.

平面 4: $y=0$, $y_x=y_z=0$

$$\iint_{S_4} z d\Omega = \int_0^1 dz \cdot \int_0^2 x dx = 1$$

$$\text{平面 } S: z=0, \quad ds = d\sigma$$

$$\iint_{S_5} y \, d\sigma = \int_0^1 dy \cdot \int_0^2 y \, dx = 1$$

$$\therefore \iint_S (y+z) \, dS = \frac{8}{3} + 2\sqrt{2}$$

$$⑦. \iint_S (x+y)^2 \, dS, \quad S \text{ 为曲面 } z = \sqrt{x^2+y^2} \text{ 及平面 } z=1 \text{ 所围之立体的表面.}$$

$$z_x = \frac{x}{\sqrt{x^2+y^2}}, \quad z_y = \frac{y}{\sqrt{x^2+y^2}},$$

$$ds = \sqrt{1 + \frac{x^2+y^2}{x^2+y^2}} \, d\sigma = \sqrt{2} \cdot d\sigma$$

$$\begin{aligned} \text{侧面: } \iint_{S_1} (x+y)^2 \, dS &= \sqrt{2} \iint_{S_1} (x+y)^2 \cdot d\sigma = \sqrt{2} \cdot \int_0^{2\pi} d\theta \int_0^1 r \cdot (r\cos\theta + r\sin\theta)^2 \cdot dr \\ &= \sqrt{2} \int_0^{2\pi} d\theta \int_0^1 r^3 (\cos\theta + \sin\theta)^2 \cdot dr \\ &= \frac{\sqrt{2}}{4} \int_0^{2\pi} (1 + 2\sin\theta\cos\theta) \cdot d\theta \\ &= \frac{\sqrt{2}}{2} \pi \end{aligned}$$

$$\text{顶面: } z=1, \quad x^2+y^2=1, \quad x=r\cos\theta, \quad y=r\sin\theta, \quad dS = d\sigma$$

$$\begin{aligned} \iint_{S_2} r^3 (\cos\theta + \sin\theta)^2 \cdot dr \cdot d\theta &= \int_0^{2\pi} (1 + 2\sin\theta\cos\theta) \cdot d\theta \cdot \int_0^1 r^3 \, dr = \frac{1}{2} \pi \\ \therefore \iint_S (x+y)^2 \, dS &= \frac{\sqrt{2}+1}{2} \pi \end{aligned}$$

$$⑧. \iint_S (x+y+z)z \, dS, \quad S \text{ 为圆锥面的一部分: } x = \rho\cos\theta\sin\alpha, y = \rho\sin\theta\sin\alpha, z = \rho\cos\alpha \quad (0 \leq \rho \leq a, 0 \leq \theta \leq 2\pi), \text{ 其中 } \alpha \in (0, \pi/2) \text{ 为常数.}$$

$$x_\rho = \cos\theta\sin\alpha, \quad y_\rho = \sin\theta\sin\alpha, \quad z_\rho = \cos\alpha$$

$$x_\theta = -\rho\sin\theta\sin\alpha, \quad y_\theta = \rho\cos\theta\sin\alpha, \quad z_\theta = 0$$

$$E=1, \quad F = -\rho\sin\theta\cos\theta\sin^2\alpha + \rho\sin\theta\cos\theta\sin^2\alpha + 0 = 0$$

$$G = \rho^2\sin^2\alpha$$

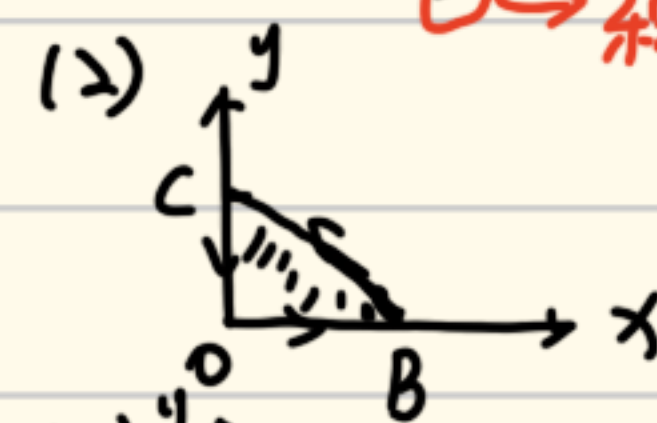
$$\therefore dS = \sqrt{\rho^2\sin^2\alpha} = \rho\sin\alpha$$

$$\begin{aligned}
 \text{原式} &= \int_0^{2\pi} d\theta \int_0^a (r - s\theta \sin\alpha + \sin\theta \sin\alpha + r - s\alpha) P \cos\alpha \cdot P \sin\alpha \, dP \\
 &= \frac{a^4}{4} \cdot \sin\alpha \cos\alpha \int_0^{2\pi} (\sin\alpha \sin\theta + \sin\alpha \cos\theta + r - s\alpha) \cdot d\theta \\
 &= \frac{a^4}{4} \cdot \sin\alpha \cos\alpha \cdot 2\pi \cdot r - s\alpha \\
 &= \frac{a^4}{2} \cdot \sin\alpha \cos^2\alpha \pi.
 \end{aligned}$$

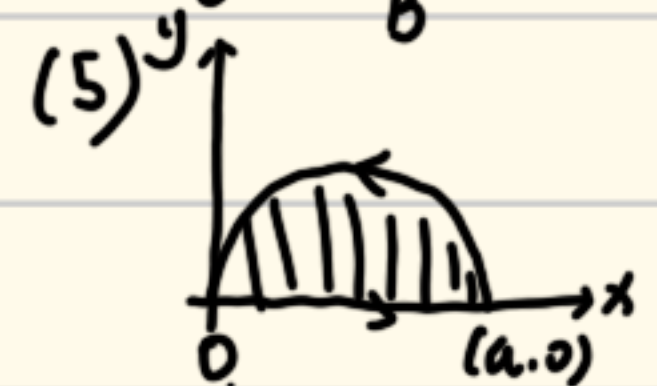
$$1. (1) \quad \frac{\partial Q}{\partial x} = y^2, \quad \frac{\partial P}{\partial y} = -x^2$$

$$\text{原式} = \iint_D (x^2 + y^2) \, d\sigma = a^2 \cdot \pi a^2 = \pi a^4$$

积分区域变化, 不能直接换, 若 L^+ 上可替换 $(1-x)$



$$\begin{aligned}
 \iint_D (2x - 2y) \, d\sigma &= 2 \int_0^1 dx \int_0^{1-x} (x - y) \, dy = 2 \int_0^1 \left[x - \frac{1}{2}(1-x)^2 \right] dx \\
 &= 2 \cdot \left(-\frac{1}{6} + 1 - \frac{1}{2} \right) = \frac{1}{3}
 \end{aligned}$$



$$\begin{aligned}
 &\iint_D (e^x \cos y - e^x \cos y + 8) \, d\sigma - \int_{L^+} (e^x \sin y + \sin x - 8y) \, dx + (e^x \cos y - \sin y) \, dy, \\
 &\quad (L^+: y=0, 0 \leq x \leq a) \\
 &= 8 \iint_D d\sigma - \int_0^a \sin x \, dx \\
 &= \pi \cdot \frac{a^2}{4} \cdot 8 - [-\cos x]_0^a = 2\pi a^2 - 1 + \cos a
 \end{aligned}$$

$$2. (1) \quad \int_L P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) \, d\sigma,$$

$$P = -y, \quad Q = x \quad S = \frac{1}{2} \int_L -y \, dx + x \, dy$$

$$dx = -3a \cos^2 t \sin t \, dt, \quad dy = 3a \sin^2 t \cos t \, dt$$

$$S = \frac{1}{2} \int_0^{2\pi} (a \sin^3 t \cdot 3a \cos^2 t \sin t + a \cos^3 t \cdot 3a \sin^2 t \cos t) \cdot dt$$

$$= \frac{3}{2} a^2 \int_0^{2\pi} (\sin^4 t \cos^2 t + \cos^4 t \sin^2 t) \cdot dt$$

$$= \frac{3}{2} a^2 \int_0^{2\pi} (\sin^4 t + \cos^4 t - \sin^6 t - \cos^6 t) \, dt$$

$$= \frac{3}{2} a^2 \cdot 4 \cdot \left(\frac{3}{8} \cdot \frac{\pi}{2} \cdot 2 - \frac{15}{48} \cdot \frac{\pi}{2} \cdot 2 \right)$$

$$= \frac{3}{2} a^2 \cdot 4 \cdot \frac{\pi}{16}$$

$$= \frac{3}{8} a^2 \pi$$

1. 应用格林公式计算下列曲线积分:

(1) $\oint_{L^+} (xy^2 + y^3) \, dy - (x^3 + x^2 y) \, dx$, 其中 L 为圆周: $x^2 + y^2 = a^2$;

(2) $\oint_{L^+} y^2 \, dx + x^2 \, dy$, 其中 L^+ 为以 $O(0,0), B(1,0), C(0,1)$ 为顶点的三角形 OBC 的正向边界线;

(3) $\oint_{L^+} \sqrt{x^2 + y^2} \, dx + y[x y + \ln(x + \sqrt{x^2 + y^2})] \, dy$, 其中 L 是以点 $A(1,1), B(2,2)$ 和 $C(1,3)$ 为顶点的三角形的正向边界线;

(4) $\oint_{L^+} (x + e^x \sin y) \, dx + (x + e^x \cos y) \, dy$, 其中 L 是双纽线 $r^2 = \cos 2\theta$ 的右半支;

(5) $\int_{L^+} (e^x \sin y + \sin x - 8y) \, dx + (e^x \cos y - \sin y) \, dy$, 其中 L 为上半圆 $0 \leq y \leq$

$\sqrt{ax - x^2}$ ($0 \leq x \leq a$) 的边界.

2. 利用曲线积分计算下列闭曲线所围图形的面积:

(1) 星形线 $x = a \cos^3 t, y = a \sin^3 t$ ($a > 0, 0 \leq t \leq 2\pi$);

(2) 心脏线

$$\begin{cases} x = a(1 - \cos t) \cos t, \\ y = a(1 - \cos t) \sin t, \end{cases} \quad 0 \leq t \leq 2\pi.$$

3. 证明: $\oint_{L^+} f(xy)(ydx + xdy) = 0$, 其中 $f(u)$ 有连续的一阶导数, L 为光滑曲线.

4. 证明下列曲线积分与路径无关, 并求积分值:

(1) $\int_{(0,0)}^{(1,1)} (x+y)dx + (x-y)dy$;

(2) $\int_{(a_1,b_1)}^{(a_2,b_2)} xy(1+y)dx + x^2\left(\frac{1}{2}+y\right)dy$;

(3) $\int_{(0,0)}^{(a,b)} e^x \cos y dx - e^x \sin y dy$.

5. 求 $\int_{\widehat{AB}} (x^4 + 4xy^3)dx + (6x^2y^2 - 5y^4)dy$ 的值, 其中 $A(-2, -1), B(3, 0)$, $\widehat{AB}$ 为任意的路径.

6. 求满足下列等式的函数 $u(x, y)$:

(1) $du = (x^2 + 2xy - y^2)dx + (x^2 - 2xy - y^2)dy$;

(2) $du = (2x \cos y - y^2 \sin x)dx + (2y \cos x - x^2 \sin y)dy$.

3. $P = y f(xy), Q = x f(xy)$

$\frac{\partial P}{\partial y} = f(xy) + xy \cdot f'(xy), \frac{\partial Q}{\partial x} = f(xy) + xy f'(xy)$

由格林公式 原式 = $\iint_D 0 d\sigma = 0$

4. (1) $\frac{\partial Q}{\partial x} = 1 = \frac{\partial P}{\partial y}$

$\int (x+y)dx = \frac{1}{2}x^2 + xy + \psi(y)$,

$x + \psi'(y) = 0 \Rightarrow \psi'(y) = -y, \psi(y) = -\frac{1}{2}y^2$

$u(x, y) = \frac{1}{2}x^2 + xy - \frac{1}{2}y^2$

原式 = $u(1, 1) - u(0, 0) = 1 - 0 = 1$

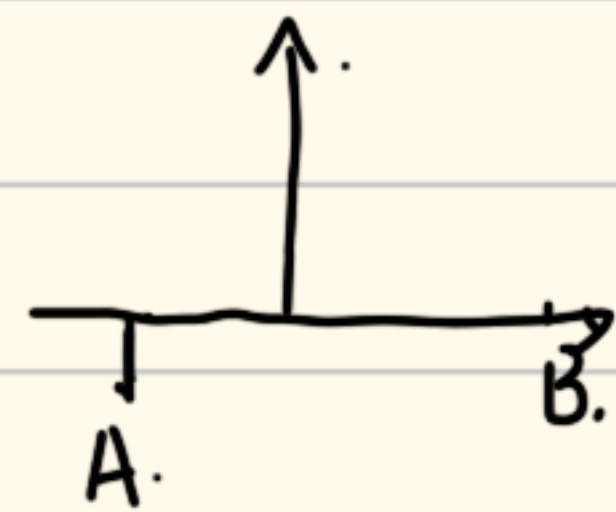
(2) $\frac{\partial Q}{\partial x} = x + 2xy = \frac{\partial P}{\partial y}$

$\int P dx = \frac{1}{2}x^2y + \frac{1}{2}x^2y^2 + C, u(x, y) = \frac{1}{2}x^2y + \frac{1}{2}x^2y^2 + \varphi(y)$

$\frac{\partial u}{\partial y} = \frac{1}{2}x^2 + x^2y + \varphi'(y) = \frac{1}{2}x^2 + x^2y, \varphi'(y) = 0$

$u(x, y) = \frac{1}{2}x^2y + \frac{1}{2}x^2y^2, u(a_2, b_2) - u(a_1, b_1) = \frac{1}{2}a_2^2b_2(1+b_2) - \frac{1}{2}a_1^2b_1(1+b_1) = \frac{1}{2}x^2y(1+y)$

5. $\frac{\partial Q}{\partial x} = 12xy^2, \frac{\partial P}{\partial y} = 12xy^2$, 与路径无关.



令 $AB: (-2, -1) \rightarrow (-2, 0) \rightarrow (3, 0)$

$L_1: (-2, -1) \rightarrow (-2, 0), x = -2, -1 \leq y \leq 0, dx = 0$

$\int_{-1}^0 (24y^2 - 5y^4) dy = 8y^3 - y^5 \Big|_{-1}^0 = -7$

$L_2: (-2, 0) \rightarrow (3, 0), y = 0, -2 \leq x \leq 3, dy = 0$

$\int_{-2}^3 x^4 dx = \frac{1}{5}x^5 \Big|_{-2}^3 = \frac{1}{5}(243 - (-32)) = \frac{275}{5} = 55$

$\therefore \text{原式} = \frac{122}{5}$

6. 求满足下列等式的函数 $u(x, y)$:

(1) $du = (x^2 + 2xy - y^2) dx + (x^2 - 2xy - y^2) dy$;

(2) $du = (2x \cos y - y^2 \sin x) dx + (2y \cos x - x^2 \sin y) dy$.

$$\int (x^2 - 2xy - y^2) dy = x^2y - xy^2 - \frac{1}{3}y^3 + C$$

$$\therefore u(x, y) = x^2y - xy^2 - \frac{1}{3}y^3 + \varphi(x)$$

$$\frac{\partial u}{\partial x} = 2xy - y^2 + \varphi'(x) = x^2 + 2xy - y^2$$

$$\varphi'(x) = x^2,$$

$$\therefore u(x, y) = \frac{1}{3}x^3 - \frac{1}{3}y^3 + x^2y - xy^2 + C$$

9. 求 $\int_{AB} (x^2 + y) dx + (x - y^2) dy$, 其中 $\widehat{AB}$ 是由 $A(0, 0)$ 至 $B(1, 1)$ 的曲线段

$$y^3 = x^2.$$

10. 设 D 是平面有界闭区域, 其边界线逐段光滑, 函数 $P(x, y), Q(x, y)$ 在 D 上有连续的一阶偏导数. 证明:

$$\oint_{L^+} [P \cos(n, x) + Q \cos(n, y)] ds = \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) d\sigma,$$

其中 $\cos(n, x), \cos(n, y)$ 为曲线 L 的外法向量的方向余弦

10 设 L 上一点正方向切向量为 $(\cos \alpha, \cos \beta)$. 则

$$\begin{cases} \cos(\vec{n}, x) = \cos \beta \\ \cos(\vec{n}, y) = -\cos \alpha \end{cases}$$

$$\text{原式} = \oint_{L^+} (P \cdot \cos \beta - Q \cos \alpha) \cdot ds$$

$$= \oint_{L^+} (-Q dx + P dy)$$

$$= \iint_D \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \right) \cdot d\sigma$$

11. 求曲线积分

$$\oint_{L^+} (x \cos \langle \vec{n}, \vec{i} \rangle + y \cos \langle \vec{n}, \vec{j} \rangle) ds,$$

其中 L 为一封闭曲线, $\vec{n}$ 为 L 的外法线方向的单位向量.

$$\text{由 10, 原式} = \iint_D 2 d\sigma = 2S.$$

S 为 L^+ 所围面积

13. 设 $u(x, y)$ 是有界闭区域 D 上的调和函数, 即 $u(x, y)$ 有连续的二阶偏导数, 且满足

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

证明:

$$(1) \oint_{L^+} u \frac{\partial u}{\partial \mathbf{n}} ds = \iint_D \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] d\sigma, \text{ 其中 } L \text{ 为 } D \text{ 的边界, } \mathbf{n} \text{ 为 } L \text{ 的外法线方向;}$$

向;

(2) 若 $u(x, y)$ 在 L 上处处为零, 则 $u(x, y)$ 在 D 上也恒为零.

(1) 设 $\vec{n} = (\cos \beta, -\cos \alpha)$, 则有 $(\cos \alpha, \cos \beta)$ 为 D 上一点正方向的单位切向量.

$$\begin{aligned} \oint_{L^+} \frac{\partial u}{\partial \vec{n}} \cdot u \, ds &= \oint_{L^+} u \cdot \left(\frac{\partial u}{\partial x} \cos \beta - \frac{\partial u}{\partial y} \cos \alpha \right) \cdot ds \\ &= \oint_{L^+} -u \cdot \frac{\partial u}{\partial y} \cdot dx + u \cdot \frac{\partial u}{\partial x} \cdot dy \\ &= \iint_D \left(\frac{\partial u}{\partial x} \cdot \frac{\partial u}{\partial x} + u \cdot \frac{\partial^2 u}{\partial x^2} + \frac{\partial u}{\partial y} \cdot \frac{\partial u}{\partial y} + u \cdot \frac{\partial^2 u}{\partial y^2} \right) d\sigma \\ &= \iint_D \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] d\sigma \end{aligned}$$

$$(2) \iint_D \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] d\sigma = 0,$$

$$\text{下面证 } \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} \equiv 0$$

假设存在一点 $(x, y) \in D$, 使 $\frac{\partial u}{\partial x} \neq 0$, 则存在一个充分小的邻域 D' , 使 $\frac{\partial u}{\partial x} \neq 0$

$$\iint_D \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \right] d\sigma \geq \iint_D \left(\frac{\partial u}{\partial x} \right)^2 d\sigma \geq \iint_{D'} \left(\frac{\partial u}{\partial x} \right)^2 d\sigma > 0, \text{ 不成立}$$

$$\text{因此 } \frac{\partial u}{\partial x} \equiv 0, \text{ 同理 } \frac{\partial u}{\partial y} \equiv 0$$

设 $u(x, y) = C$ 为常数,

而边界 $L \subset D$ 在 L 上有 $u(x, y) = 0$, 则 $C = 0$

即在 D 上 $u(x, y) \equiv 0$

求下列第一型曲面积分：

1. $\iint_S \left(2x + \frac{4}{3}y + z\right) dS$, S 为平面 $\frac{x}{2} + \frac{y}{3} + \frac{z}{4} = 1$ 在第一卦限中的部分.

2. $\iint_S (y+z) dS$, 其中 S 为平面 $y+z=1, x=2$, 以及三个坐标平面所围成之立体的表面.

3. $\iint_S (x+y+z) dS$, S 为上半球面 $z = \sqrt{a^2 - x^2 - y^2}$.

4. $\oiint_S (xy+z)^2 dS$, S 为球面 $x^2 + y^2 + z^2 = R^2$.

5. $\iint_S (x+y)^2 dS$, S 为曲面 $z = \sqrt{x^2 + y^2}$ 及平面 $z=1$ 所围之立体的表面.

6. $\iint_S [x(y+z) + z(x+y)] dS$, S 为圆锥面 $z = \sqrt{x^2 + y^2}$ 被曲面 $x^2 + y^2 = 2ax$ ($a > 0$) 所割下的部分.

7. $\iint_S yz dS$, S 为螺旋曲面

$$x = u \cos v, \quad y = u \sin v, \quad z = v \quad (0 \leq u \leq a, 0 \leq v \leq 2\pi).$$

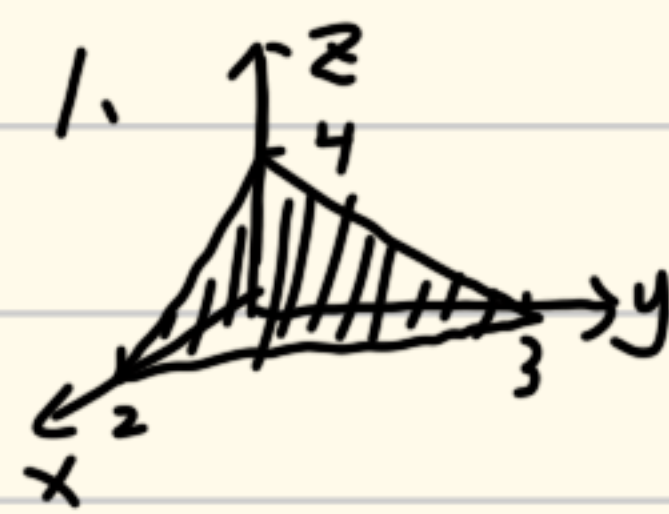
8. $\iint_S (x+y+z)z dS$, S 为圆锥面的一部分: $x = \rho \cos \theta \sin \alpha, y = \rho \sin \theta \sin \alpha, z = \rho \cos \alpha$ ($0 \leq \rho \leq a, 0 \leq \theta \leq 2\pi$), 其中 $\alpha \in (0, \pi/2)$ 为常数.

9. 求抛物面壳 $z = \frac{1}{2}(x^2 + y^2)$ ($0 \leq z \leq 1$) 的质量, 其面密度为 $\rho = x + y + z$.

10. 求面密度为 ρ_0 的均匀半球壳 $x^2 + y^2 + z^2 = a^2$ ($z \geq 0$) 对 Oz 轴的转动惯量.

11. 求一均匀圆柱面 $x^2 + y^2 = R^2$ ($0 \leq z \leq h$) 对位于 $(0, 0, h)$ 处一单位质点的引力 (面密度为常数 1).

12. 求位于第一卦限中的球面 $x^2 + y^2 + z^2 = a^2$ 的质心坐标, 设该球面的面密度为常数.



$$6x + 4y + 3z = 12.$$

$$z = 4 - 2x - \frac{4}{3}y$$

$$dS = \sqrt{1 + 4 + \frac{16}{9}} = \frac{\sqrt{61}}{3}$$

$$\therefore \frac{\sqrt{61}}{3} \iint_D 4 \cdot d\phi = 4\sqrt{61}$$

3. $x = a \sin \varphi \cos \theta, \quad y = a \sin \varphi \sin \theta, \quad z = a \cos \varphi$

$$x_\varphi = a \cos \varphi \cos \theta, \quad y_\varphi = a \cos \varphi \sin \theta, \quad z_\varphi = -a \sin \varphi$$

$$x_\theta = -a \sin \varphi \sin \theta, \quad y_\theta = a \sin \varphi \cos \theta, \quad z_\theta = 0$$

$$E = a^2, \quad G = a^2 \sin^2 \varphi, \quad F = -a \sin \varphi \cos \varphi \sin \theta \cos \theta + a^2 \sin \varphi \cos \varphi \sin \theta \cos \theta = 0$$

$$dS = \sqrt{EG - F^2} = a^2 \sin \varphi d\phi$$

$$\begin{aligned} & \iint_S (a \sin \varphi (\sin \theta + \cos \theta) + a \cos \varphi) \cdot a^2 \sin \varphi d\phi \\ &= a^3 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} [\sin \varphi (\sin \theta + \cos \theta) + \cos \varphi] \sin \varphi d\varphi \end{aligned}$$

$$= a^3 \int_0^{2\pi} \left[\frac{2\pi}{4} \cdot (\sin \theta + \cos \theta) + \frac{1}{2} \right] d\theta$$

$$= a^3 \pi.$$

$$-\frac{1}{3} \cos^3 \varphi$$

$$\frac{1}{4} - \frac{1}{4}$$

4. $x^2 y^2 + 2xy z + z^2$, 由对称性

$$\iint_S x^2 y^2 + z^2 dS,$$

$$\iint_S R^4 \sin^4 \varphi \sin^2 \theta \cos^2 \theta + R^2 \cos^2 \varphi d\varphi d\theta$$

$$= R^2 \iint_S R^2 \sin^4 \varphi \sin^2 \theta \cos^2 \theta + \cos^2 \varphi d\varphi d\theta$$

$$= 2R^2 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} R^2 \sin^4 \varphi \sin^2 \theta \cos^2 \theta + \cos^2 \varphi d\varphi$$

$$= 2R^2 \int_0^{2\pi} \left(\frac{3}{16} \pi R^2 \sin^2 \theta \cos^2 \theta + \frac{2\pi}{4} \right) d\theta$$

$$= 2R^2 \int_0^{2\pi} \left[\frac{3}{16} \pi R^2 (\sin^2 \theta - \sin^4 \theta) + \frac{2\pi}{4} \right] d\theta$$

$$= 2R^2 \left(\frac{3}{64} \pi^2 R^2 + \frac{2\pi}{2} \right)$$

$$= \frac{3}{32} \pi^2 R^4 + \pi^2 R^2$$

$$\iint_S R^2 \sin \varphi \cdot (R^4 \sin^4 \varphi \sin^2 \theta \cos^2 \theta + R^2 \cos^2 \varphi) d\varphi d\theta$$

$$= 2R^4 \int_0^{2\pi} d\theta \int_0^{\frac{\pi}{2}} R^2 \sin^5 \varphi \sin^2 \theta \cos^2 \theta + \cos^2 \varphi \sin \varphi d\varphi$$

$$= 2R^4 \int_0^{2\pi} \left(\frac{8}{15} \sin^2 \theta \cos^2 \theta R^2 + \frac{1}{3} \right) d\theta$$

$$= 2R^4 \left(\frac{8}{15} \times 4 \times \left(\frac{2\pi}{4} - \frac{3\pi}{8} \right) + \frac{4\pi}{3} \right)$$

$$= \frac{4}{15} \pi^2 R^6 + \frac{4\pi}{3} R^4$$

$$6. \quad x=r\cos\theta, \quad y=r\sin\theta, \quad z=r.$$

$$dS = \sqrt{EG-F^2}d\theta, \quad E=2, G=r^2, F=\cos\theta \cdot -r\sin\theta + \sin\theta \cdot r\cos\theta + 0 = 0$$

$$dS = \sqrt{2} r \cdot d\theta$$

$$r \leq 2a\cos\theta$$

$$\iint r^2 \sin\theta \cos\theta + 2r^2 \cos\theta + r^2 \sin\theta \, d\theta dr.$$

$$= \int_0^{22} d\theta \int_0^{2a\cos\theta} r^2 (\sin\theta \cos\theta + 2\cos\theta + \sin\theta) dr$$

$$= \frac{8a^3}{3} \int_0^{22} (\sin\theta \cos^4\theta + 2\cos^4\theta + \sin\theta \cos^3\theta) d\theta$$

$$= \frac{8a^3}{3} \cdot 2 \cdot \frac{2}{2} \cdot \frac{3}{8}$$

$$= 4a^3$$

$$\sqrt{2} \iint r^3 (\sin\theta \cos\theta + 2\cos\theta + \sin\theta) d\theta dr.$$

$$= \sqrt{2} \int_0^{22} d\theta \int_0^{2a\cos\theta} r^3 \dots dr$$

$$= 4\sqrt{2} a^4 \int_0^{22} \cos^4\theta (\sin\theta \cos\theta + 2\cos\theta + \sin\theta) d\theta$$

$$= 4\sqrt{2} a^4 \cdot 4 \cdot \frac{8}{15}$$

$$= \frac{128\sqrt{2}}{15} a^4$$

$$6. \iint_S [x(y+z) + z(x+y)] dS, \quad S \text{ 为圆锥面 } z = \sqrt{x^2+y^2} \text{ 被曲面 } x^2+y^2=2ax$$

($a > 0$) 所割下的部分.

$$7. \iint_S yz dS, \quad S \text{ 为螺旋曲面}$$

$$x = u\cos v, \quad y = u\sin v, \quad z = v \quad (0 \leq u \leq a, 0 \leq v \leq 2\pi).$$

$$8. \iint_S (x+y+z)z dS, \quad S \text{ 为圆锥面的一部分: } x = \rho\cos\theta\sin\alpha, y = \rho\sin\theta\sin\alpha, z =$$

$$\rho\cos\alpha \quad (0 \leq \rho \leq a, 0 \leq \theta \leq 2\pi), \text{ 其中 } \alpha \in (0, \pi/2) \text{ 为常数.}$$

1

$$7. \quad x_u = \cos v, \quad y_u = \sin v, \quad z_u = 0$$

$$x_v = -u\sin v, \quad y_v = u\cos v, \quad z_v = 1$$

$$dS = \sqrt{u^2+1} d\theta$$

$$\iint_S uv \sin v \sqrt{u^2+1} d\theta = \int_0^{22} dv \int_0^a u \sqrt{u^2+1} \cdot v \sin v du = \left(\frac{1}{3}(u^2+1)^{\frac{3}{2}} - \frac{1}{3} \right) \cdot \int_0^{22} v \sin v dv = \frac{22}{3} \cdot (1 - (a^2+1)^{\frac{3}{2}})$$

$$\frac{1}{2} \int_0^a \sqrt{u^2+1} \cdot du^2 = \frac{1}{4} \frac{2}{3} \cdot (u^2+1)^{\frac{3}{2}} \Big|_0^a = \frac{1}{3} (a^2+1)^{\frac{3}{2}} - \frac{1}{3}$$

$$-v \cos v = -[v \cos v]_0^{22} + \int_0^{22} \cos v dv$$

$$= -(22)$$

$$9. \text{ 求抛物面壳 } z = \frac{1}{2}(x^2+y^2) \quad (0 \leq z \leq 1) \text{ 的质量, 其面密度为 } \rho = x+y+z.$$

$$10. \text{ 求面密度为 } \rho_0 \text{ 的均匀半球壳 } x^2+y^2+z^2=a^2 \quad (z \geq 0) \text{ 对 } Oz \text{ 轴的转动惯量.}$$

$$11. \text{ 求一均匀圆柱面 } x^2+y^2=R^2 \quad (0 \leq z \leq h) \text{ 对位于 } (0,0,h) \text{ 处一单位质点的引力 (面密度为常数 1).}$$

$$12. \text{ 求位于第一卦限中的球面 } x^2+y^2+z^2=a^2 \text{ 的质心坐标, 设该球面的面密度为常数.}$$

$$9. \quad 0 \leq x^2+y^2 \leq 2, \quad \text{由对称性, } \iint (x+y) dS = 0$$

$$x = r\cos\theta, \quad y = r\sin\theta, \quad z = \frac{1}{2}r^2$$

$$x_r = \cos\theta, \quad y_r = \sin\theta, \quad z_r = r.$$

$$x_\theta = -r\sin\theta, \quad y_\theta = r\cos\theta, \quad z_\theta = 0$$

$$dS = \sqrt{(r^2+1)r^2} d\theta = r \sqrt{r^2+1} d\theta$$

$$4 \int_0^{\frac{\pi}{2}} d\theta \int_0^{\sqrt{2}} \frac{1}{2} r^3 \sqrt{r^2+1} dr = 2 \int_0^{\sqrt{2}} r^3 \sqrt{r^2+1} dr, \quad r = \tan\theta$$

$$\begin{aligned}
 & \int_0^{\sqrt{3}} r^3 \sqrt{r^2+1} dr, \quad r = \tan \theta, \\
 & \int_0^{\arctan \sqrt{3}} \tan^3 \theta \cdot \sec^3 \theta \cdot d\theta \\
 & = \int_0^{\arctan \sqrt{3}} (\sec^2 \theta - 1) \cdot \sec^2 \theta d\sec \theta \\
 & = \int_0^{\arctan \sqrt{3}} \sec^4 \theta - \sec^2 \theta d\sec \theta \\
 & = \left. \frac{1}{5} \sec^5 \theta - \frac{1}{3} \sec^3 \theta \right|_0^{\arctan \sqrt{3}} \\
 & = \frac{9\sqrt{3}}{5} - \sqrt{3} + \frac{2}{15} \\
 & = \left(\frac{4\sqrt{3}}{5} + \frac{2}{15} \right) 2
 \end{aligned}$$

$$= 2 \cdot \int_0^{\arctan \sqrt{3}} \tan^3 \theta \cdot \sec \theta d\theta$$

$$= 2 \cdot \int_0^{\arctan \sqrt{3}} \tan^2 \theta \cdot d\sec \theta$$

$$= 2 \cdot \int_0^{\arctan \sqrt{3}} (\sec^2 \theta - 1) d\sec \theta$$

$$= 2 \cdot \left(\frac{1}{3} \sec^3 \theta - \sec \theta \right) \Big|_0^{\arctan \sqrt{3}}$$

$$= 2 \cdot \left(\frac{3\sqrt{3}}{3} - \sqrt{3} \right) = \frac{2}{3} 2$$

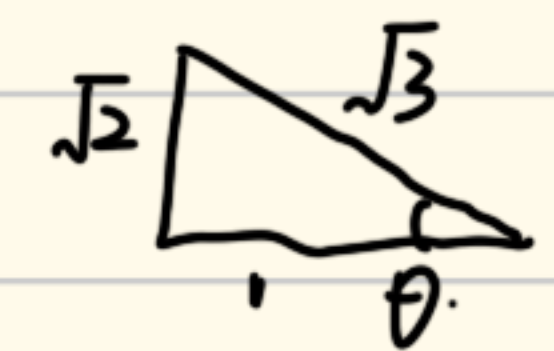
$$\lim_{\theta \rightarrow 0} \left(\frac{1}{3 \cos^3 \theta} - \frac{1}{\cos \theta} \right)$$

$$= \lim_{\theta \rightarrow 0} \frac{1 - 3 \cos^2 \theta}{3 \cos^3 \theta} = -\frac{2}{3}$$

$$(\tan \theta)' = \sec^2 \theta$$

$$(\sec \theta)' = \frac{\sin \theta}{\cos^2 \theta} = \tan \theta \cdot \sec \theta$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$



$$\sec \theta = \sqrt{2}$$

